

PID No.

9th Edition TECHgium®

Presentation Round

Concept Title

Purposeful.
Agile.
Innovation.

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Name 4

College name, City, State

Challenge Statement

Bringing Healthcare to Everyone: Real-Time Monitoring and AI Solutions

Statements

Design an AI-powered healthcare solution that enables real-time patient monitoring and prioritization. The system should include a wearable device to track vital signs and detect early indicators of health issues, a remote dashboard for clinicians to monitor patients, and an AI model that helps prioritize care based on urgency. The solution must be cost-effective, scalable, and suitable for deployment in rural and resource-constrained settings.

Background

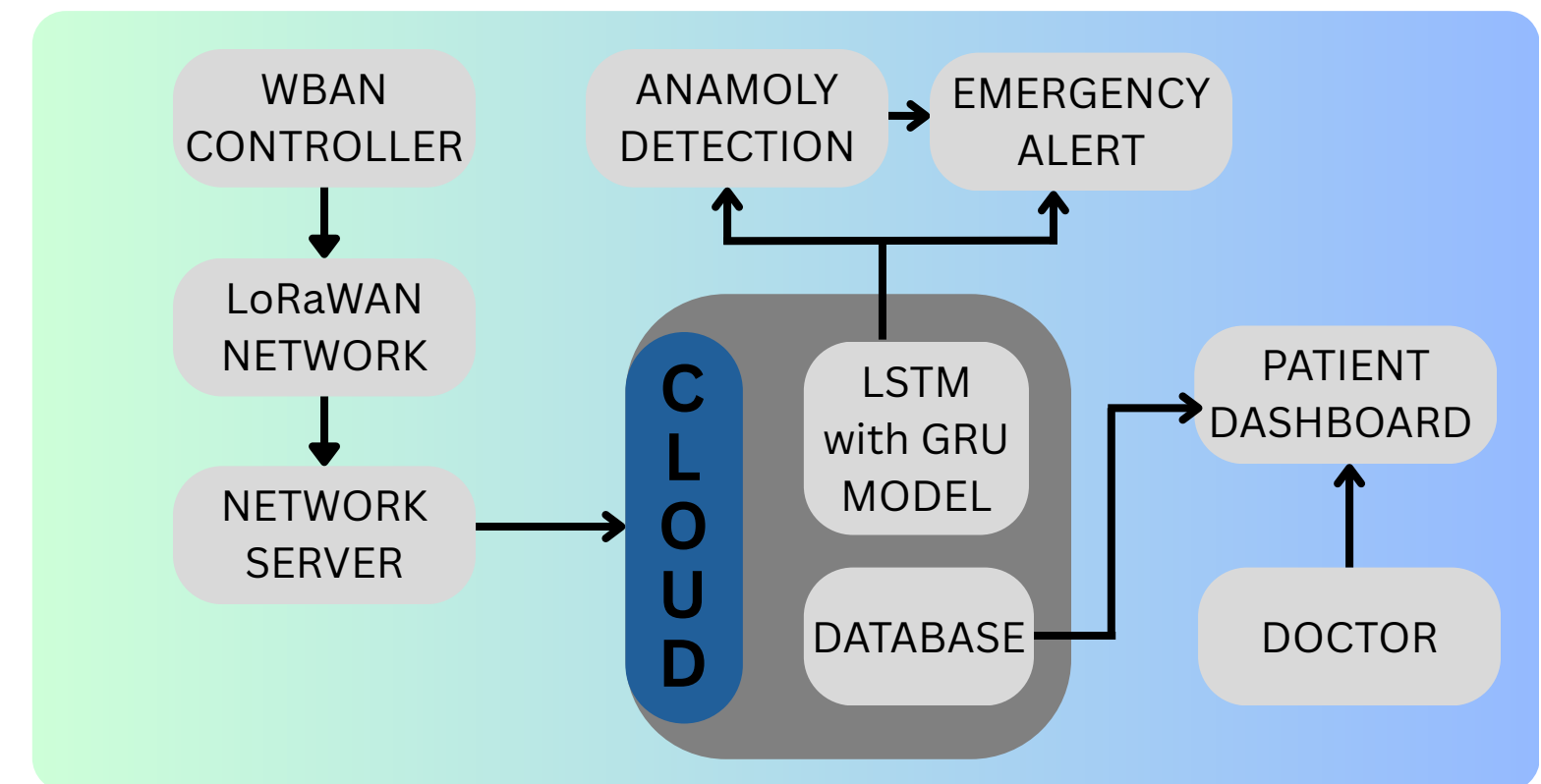
Many healthcare systems, especially in remote and underserved areas, struggle with real-time patient monitoring due to limited resources and infrastructure. This lack of timely data can delay diagnosis and treatment, increasing health risks. There is a growing need for affordable, scalable solutions that leverage wearable technology and AI to improve access to care and enable proactive health management.

Deliverables

A prototype of a wearable monitoring device, a user-friendly dashboard for remote patient tracking, and an AI model that supports patient triage based on health data. The solution should demonstrate its potential to improve healthcare accessibility, reduce response times, and support data-driven decision-making in clinical settings.

Proposed Solution

- We propose a deployable **Wireless Body Area Network (WBAN)** solution in which **patients** are **equipped with wearable sensors** that **capture the vital parameters** continuously.
- The **vitals extracted** separately will be **aggregated** in a microcontroller and that performs the **lightweight pre-processing** and produce the **summarized health data**.
- The **summarized health data** with a uniquely assigned device identifier (patient ID), will be Transmitted to the **LoRaWAN network server(LNS)** through a low-bandwidth **LoRaWAN network infrastructure**.
- From **LNS** the data will be **sent to cloud** through internet and will be **stored** in a database and also delivers an **AI-driven monitoring service** that continuously analyzes incoming data and **flags potential life-threatening anomalies** at the **earliest possible stage**.
- A **web-based dashboard** enables hospitals to **onboard and manage patients** within the network, **review aggregated health reports**, and **track network-related costs** incurred for each patient.



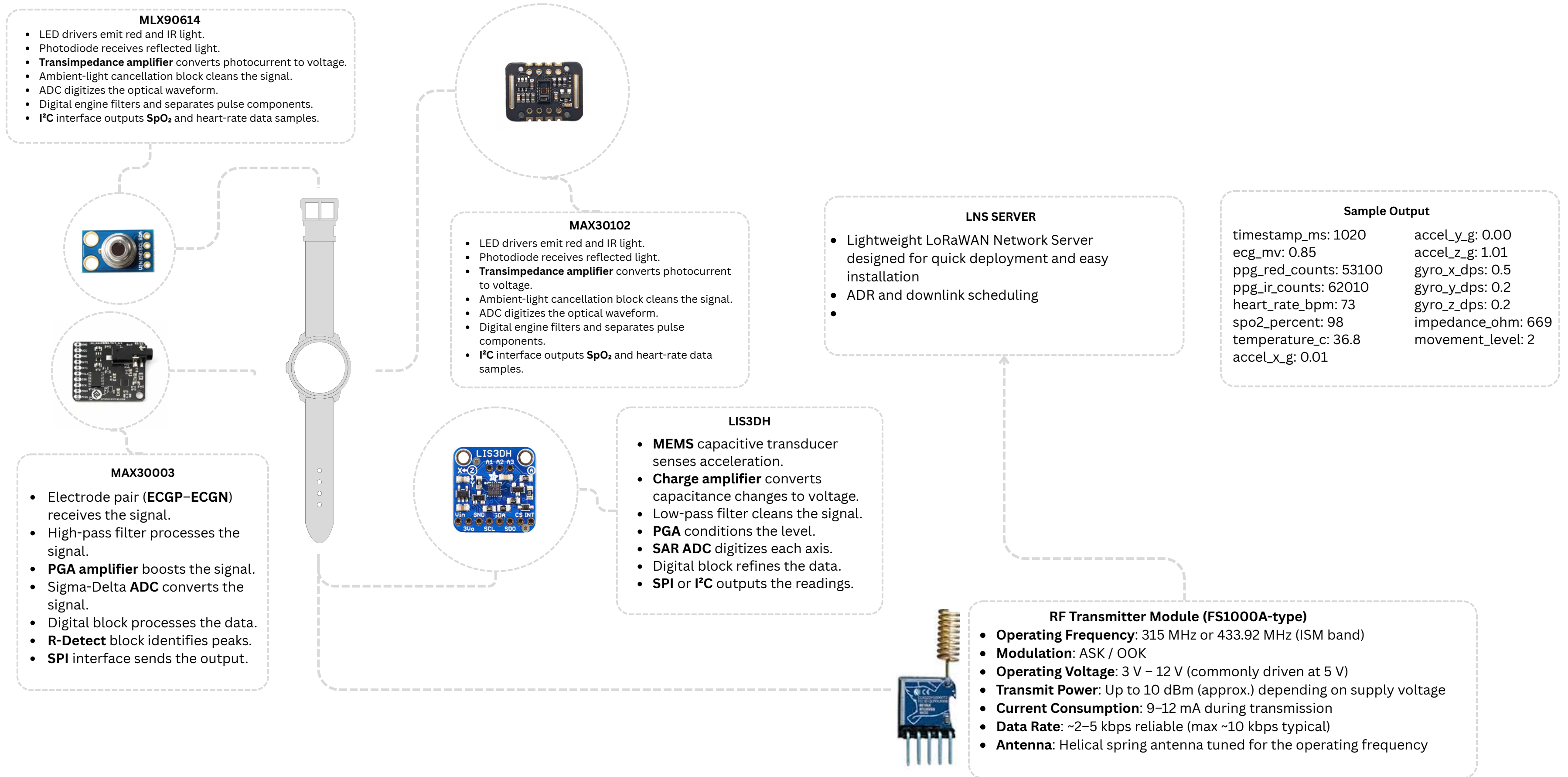
CHALLENGES SOLVED

- Continuous vital-sign acquisition and transmission in regions with limited or no internet connectivity
- Solving operational inefficiencies for hospitals by providing a unified dashboard for patient onboarding, real-time monitoring, report aggregation

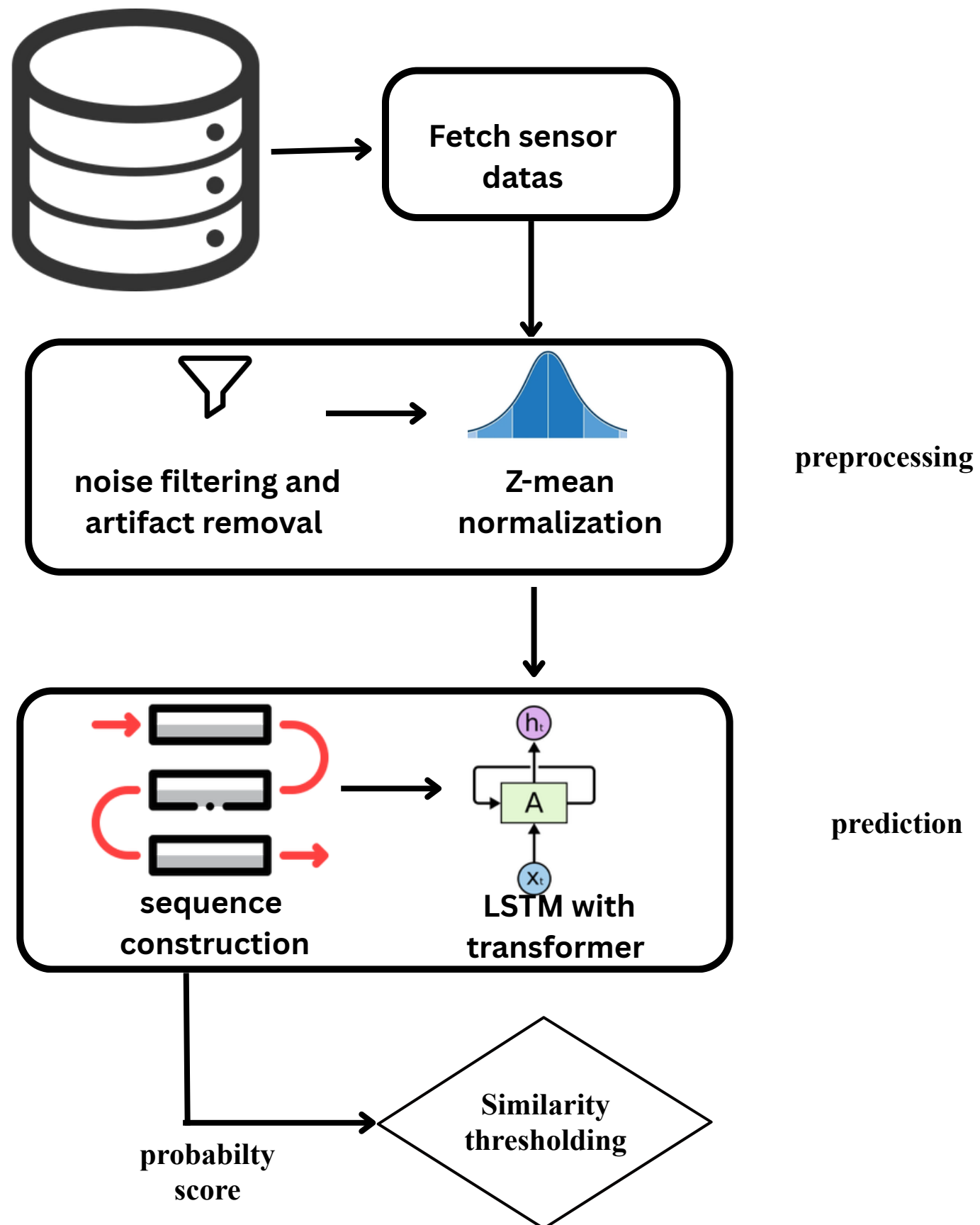
Pros and Cons of the solution

	Pros		
	Smart Wearable for Continuous Health Tracking A simple, lightweight wearable that quietly measures key body signals all day without disturbing the user.		Limited battery capacity requires regular charging.
	Automated Alerts and Early-Warning Signals The system detects unusual patterns and flags them early, helping prevent emergencies before they escalate.		Sensor accuracy can degrade with poor skin contact or motion.
	Data-Driven Decision Support for Faster Care Clear visual cues and summaries that help medical staff make quicker and more informed choices.		Continuous monitoring increases data volume and storage demands.
	AI-Based Patient Prioritization An AI engine that sorts patients by urgency so clinicians know exactly who needs attention first.		AI prioritization may mis-rank cases if training data is insufficient.
	Community Health Worker Integration Local volunteers or ASHA workers can help with device setup, charging, and patient checks.		Rural offline operation delays real-time clinical decisions.
	Offline-Friendly Support for Rural Areas Designed to work even with weak or unstable internet by storing and forwarding data intelligently.		Low-cost components may reduce durability in harsh environments.
	Easy-to-Use Remote Monitoring Dashboard A clean, simple interface where doctors and health workers can view patient updates in real time.		Dashboard usability depends on stable device–server synchronization.

Technical Description



Technical Description



- **Pre processing stage:** Vitals are collected every 5 minutes and preprocessed using **artifact rejection**, and **z-score normalization**, while missing data is managed via **forward-filling** or **sentinel values** to ensure a **continuous timeline**.
- **Hybrid LSTM-Transformer Model:** A rolling 5-hour window of data is processed by a hybrid architecture, where an **LSTM network extracts short-term temporal trends** and a **Transformer Encoder captures long-range dependencies** and complex interactions.
- **Risk Stratification & Alerting:** The system outputs a calibrated **probability of deterioration** using which **thresholding** is made in order to classify the patient as urgent and non urgent cases

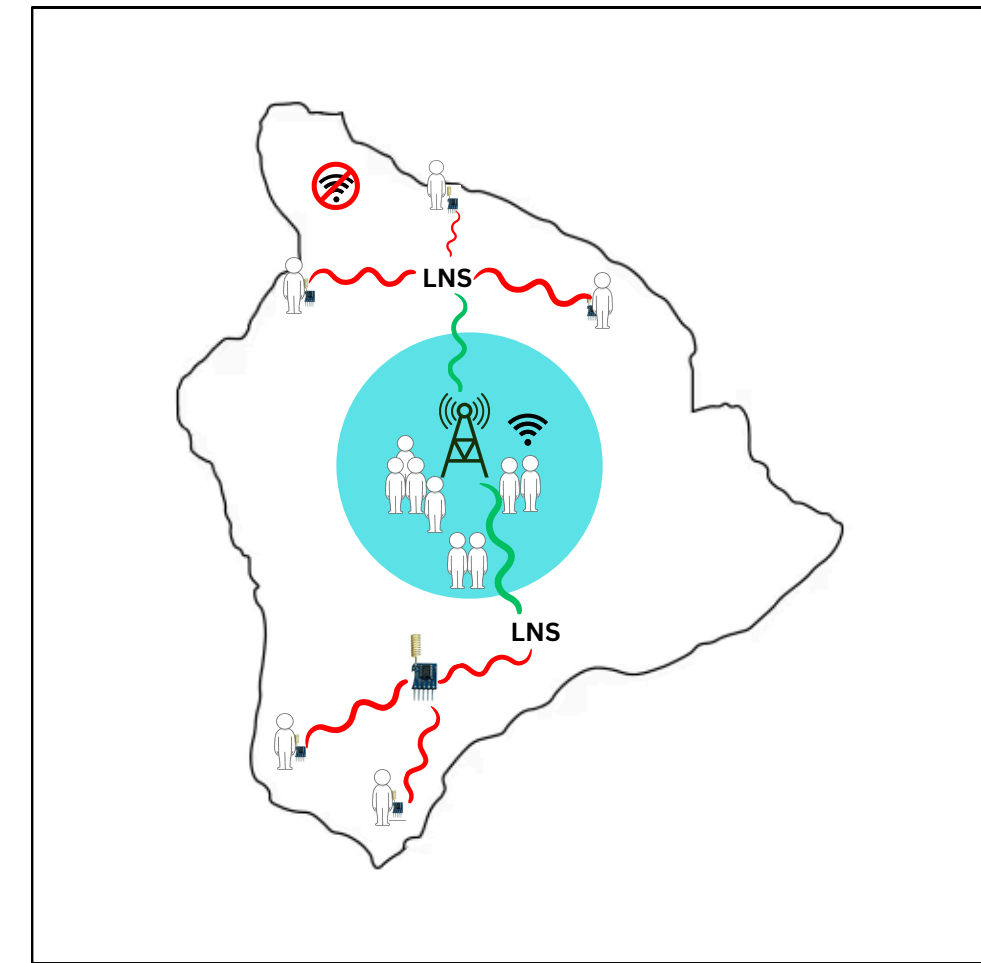
Implementation Plan

PHASE - I MINIMUM VIABLE PRODUCT

- Each patient will be provided with a wearable device (sensor nodes + microcontroller) for vital monitoring summarizing the health data.
- A scalable cloud infrastructure will be established to manage and store high-volume physiological data.
- The microcontroller that can communicate to smart phones can send summarized health data to the cloud infrastructure built.
- A web platform will be developed to support patient onboarding, administration, and ongoing management and displaying aggregated patient dashboard.

PHASE - II EXTENSION FOR NETWORK CONSTRAINED REGIONS

- The patient will receive an enhanced wearable device that incorporates a LoRaWAN communication module.
- The LNS will be strategically installed to provide optimal coverage and clear line-of-sight to the LoRaWAN end nodes (with respect to surveys).
- The LNS will manage all network communication, data routing, and connectivity for the deployed devices.
- Multi hop LNS can also be integrated if necessary.



PHASE - III AI MODULE EXTENSION

- The patient will receive an enhanced wearable device that incorporates a LoRaWAN communication module.
- The LNS will be strategically installed to provide optimal coverage and clear line-of-sight to the LoRaWAN end nodes (with respect to surveys).
- The LNS will manage all network communication, data routing, and connectivity for the deployed devices.
- Multi hop LNS can also be integrated if necessary.

Implementation Plan

- **. Establish a Strong Baseline Using Research Data:**We begin by training the initial model on the published reference dataset. This baseline is enhanced by engineering additional features such as HRV, entropy, and trend-based indicators to improve predictive performance from the outset.
- **2. Standardize the Data Processing Pipeline:**All physiological signals are normalized and organized into consistent 5-hour sliding windows. This ensures the model receives a uniform, high-quality historical input sequence with no data leakage.
- **3. Train the Core LSTM Model with Attention:**An transformer with LSTM architecture is trained to integrate vital signs, motion context, and derived features. The training process incorporates class-imbalance handling to reliably detect rare but critical deterioration events.
- **4. Pilot Deployment for Local Data Collection:**The model is first deployed in a limited geographic region to validate real-world performance and collect region-specific physiological and environmental data unique to the target population.
- **5. Iterative Retraining and Scalable Deployment:**Using the newly collected local data, the model is periodically retrained to improve accuracy and adaptability. Once validated, the refined model is scaled to larger regions, forming a continuous improvement loop over time.

Validation / Testing / Analysis

Cost Estimate

[illegible]

Assumptions

- The LoRaWAN network has to be deployed on the regions where the bandwidth conditions not met.
- The hospitals will subscribe to the network laid and can able to utilize it by allocating patients to the network
- On the regions without network constraints , the central node of the sensor architecture will directly communicate with the cloud through a intermediate node such as smart phones , smart watches etc.. .

References

- Development and validation of a clinical. wearable deep learning based continuous inhospital deterioration prediction model
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Thank You

Guidelines

- You can make minor changes but do not use more than 15 slides.
- Do not give any theoretical fundamentals.
- Illustrate with suitable means such as drawings, examples, algorithms, infographics etc.
- Use a consistent format. Do not change the provided template.
- Check for spelling and grammar errors.
- Highlight the major advantages of the technique/method proposed when compared with other techniques or conventional techniques.
- While uploading the PPT file, rename first 4 letters (i.e., 0000) with your PID nos.

